Laboratory 7

Investigating thermal conductivity and convection rates for the Lumped Capacitance Model.

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MAE 170

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MAE 170 Lab 7

Abstract

Understanding how heat transfers across a material through convection is of vital importance when concerned with maintaining or controlling temperature. This study has applicability in a plethora of engineering applications, ranging from cooling electronics in technology to maintaining structural integrity in extreme temperature systems like rocket engines. This experiment studies material properties, like heat transfer coefficients, and their influence on the cooling behavior of thin plate heat-sinks made from aluminum and acrylic. Using an automated Arduino-Matlab script, acrylic and aluminum plates were heated to 35°C and cooled by free convection and forced convection processes. Forced convection utilized a fan to drive cool air over the plates, while free convection allowed the plates to cool with the natural environment. By measuring the change in temperature as each plate cooled, the convective heat transfer coefficients and cooling rates were experimentally determined for each of the four cases. Overall, Aluminum was found to have a higher convective heat transfer coefficient than acrylic, and forced convection had a faster cooling rate than free convection. This experiment demonstrated the importance of material choice and convective processes in designing systems involving thermal control.

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Introduction

Convective heat transfer is an essential mechanism in the development of modern-day energy generation, energy storage, and material processing. Utilizing it properly enables engineers to increase the efficiency and reliability of existing techniques, allowing for improvement of modern technologies. The rate at which a solid transfers heat to or from its surroundings is affected by two factors: the material's thermal properties, like thermal conductivity and density, and the convection regime of the material's environment. A material can experience either free convection, where temperature and density disparities allow for natural heat transfer between the material and the ambient surrounding medium, or forced convection induced by a fan, where an external source, like a fan, forces the fluid over the material, which forces convection to occur at a much faster rate [1].

To study heat transfer in materials through convection, scientists and engineers commonly use the Lumpled-Capacitance Model, which assumes there are no temperature gradients within a material. The model holds that the temperature within a solid remains uniform when a material is both being heated and cooled [2]. A thorough understanding of how material properties and convection regimes affect the heat transfer of a material and the scope of the Lumped-Capacitance Model is essential for engineers and scientists to be able to maximise the accuracy and reliability of their designs.

This experiment studies the effect of different material types and different convection types on the heat transfer of a square plate. Thin square plates were used to model a heat sink and assess the applicability and limitations of the Lumped-Capacitance Model. A lamp was positioned parallel to the plates to raise their temperature to 35 °C. At this point, the lamp was retracted, and the plates were allowed to cool to the ambient temperature. The temperature and time were recorded on MATLAB and used to derive experimental Biot number values for 4 cases of varied convection and material. The Biot number is a dimensionless number that compares a material's internal conduction resistance to its external convection resistance. It helps to indicate whether temperature gradients within a material are significant, which helps indicate if the Lumped-Capacitance Model is valid.

Methodology

Experimental Setup

To investigate how material type and convection type affected heat transfer rate, this experiment used an automated Arduino custom PCB system.

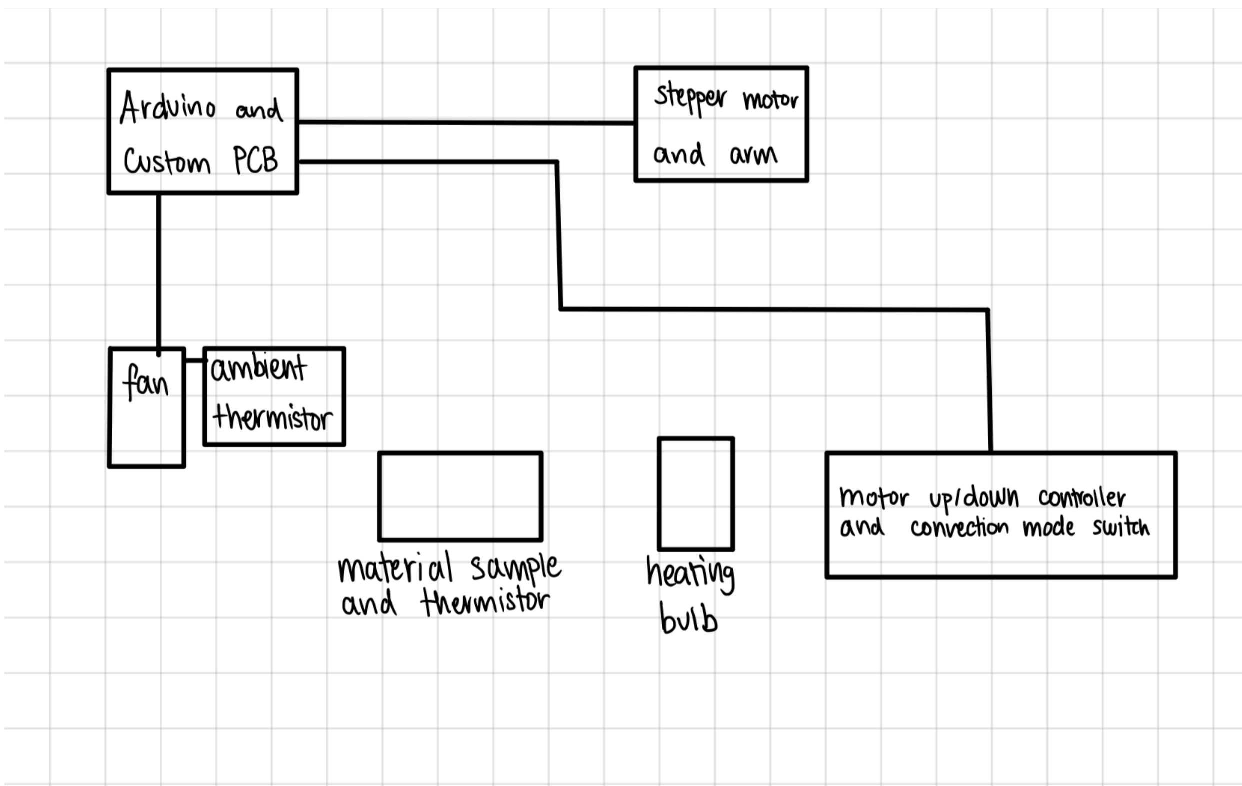

Figure A. Experimental Setup

Methods

To begin the experiment, a material was placed on the test stand, and the convection type was selected on the PCB board via a switch. For the first stage of the experiment, the Arduino commanded the stepper motor to rotate 90° counterclockwise, positioning the lamp on the motor's arm directly above the plate and its thermistor. Data was then recorded and stored in a MATLAB matrix for later analysis.

Once the detected plate's temperature reached the target of 35 degrees, the Arduino commanded the stepper motor to turn off the lamp and rotate 90 degrees clockwise to allow the plate to begin cooling. If there was a sudden down spike in temperature after the lamp was turned off, the Arduino would turn the lamp on again to briefly reheat the plate to 35 degrees before raising up.

Depending on the convection type selected, the Arduino either turned on the fan to start the forced convection process or left the fan off to start cooling by free convection. During the cooling phase, the Arduino continued to measure temperature across time using thermistor data and stored values in MATLAB. The experiment ran until the Arduino detected the plate's temperature to be within 2 degrees of the recorded ambient temperature. The experiment would then repeat until three trials were collected for each of the four cases. Data was averaged across trials, and the transient solution was used to determine the Biot number and convective heat transfer coefficient for each scenario.

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Results

Data analysis was only performed on the cooling portions of the collected MATLAB data, as this carried information about the convective properties of the samples. The temperature and time values were first manipulated using equations 1 and 2 to non-dimensionalize the data. The goal of this was to set up derivations for the Biot number and heat transfer coefficient calculations, and also improve the reproducibility of the experiment.

$$\theta = \frac{T - T_{\infty}}{T_0 - T_{\infty}} \quad (1)$$

$$\tau = \frac{kAt}{\rho c d V} \quad (2)$$

See tables D and E in the Appendix tables section for variable definitions.

An average from three trials of each case was stored in a MATLAB matrix and used for later calculations. The non-dimensionalized data from the equations above were plotted against each other for both the Aluminum and Acrylic plate scenarios, as seen in Figures 1A and 1B below.

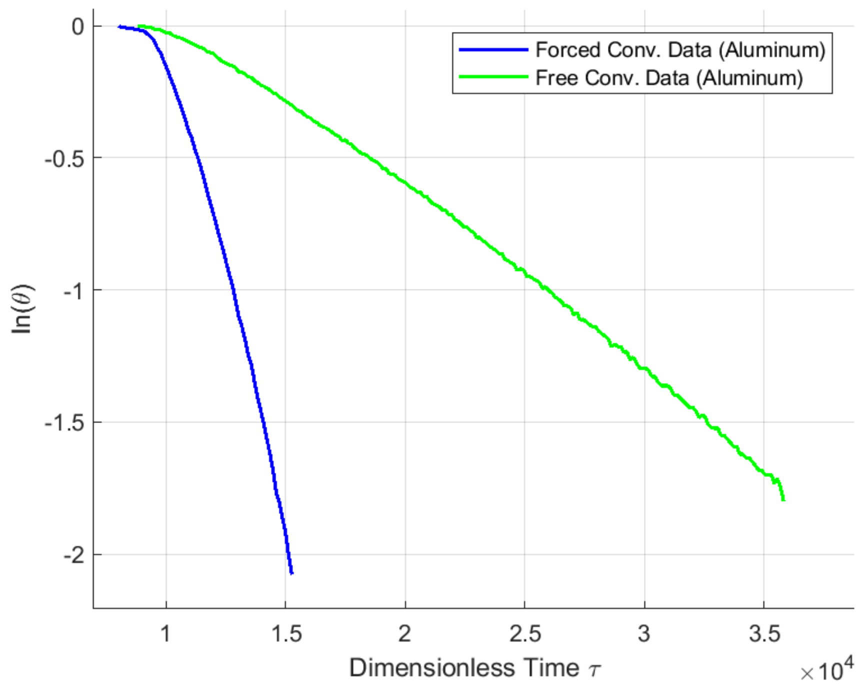

Figure 1A. *Natural Logarithm of Dimensionless temperature, $ln(\theta)$, plotted against dimensionless time spent cooling, τ. Forced convection (blue) vs. free convection (green) for an Aluminum square plate*

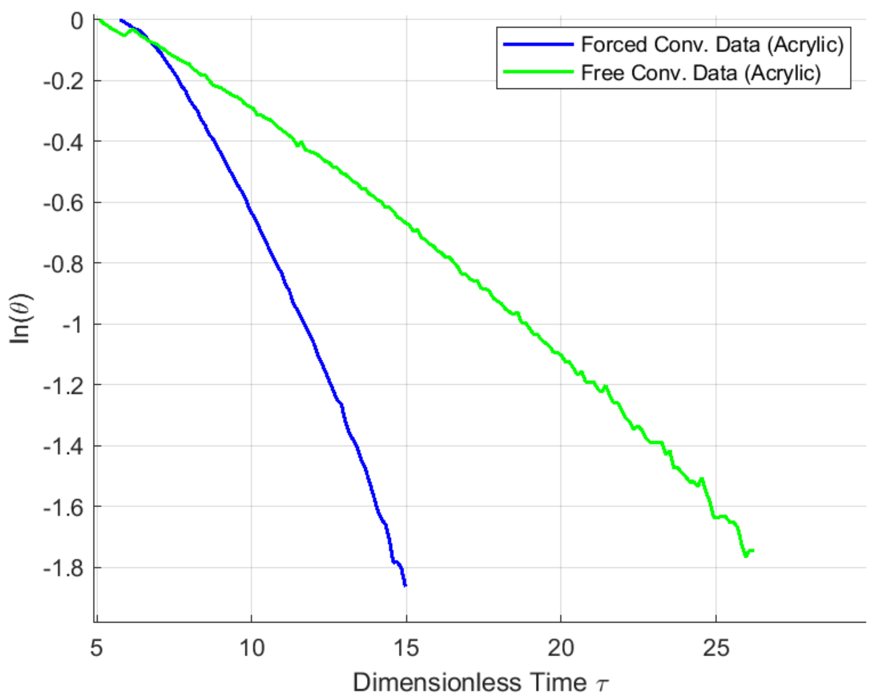

Figure 1B. *Natural Logarithm of Dimensionless temperature, $ln(\theta)$, plotted against dimensionless time spent cooling, τ. Forced convection (blue) vs. free convection (green) for an Acrylic square plate*

As pictured in the figures above, despite all scenarios having an initial temperature of 35 degrees, the curves for the Aluminum plate had much faster decay for both free and forced convection compared to the Acrylic curves.

By using a linear regression and MATLAB's polyfit function, a line of best fit was fitted to each of the four cases, as shown in Figures 2A and 2B below.

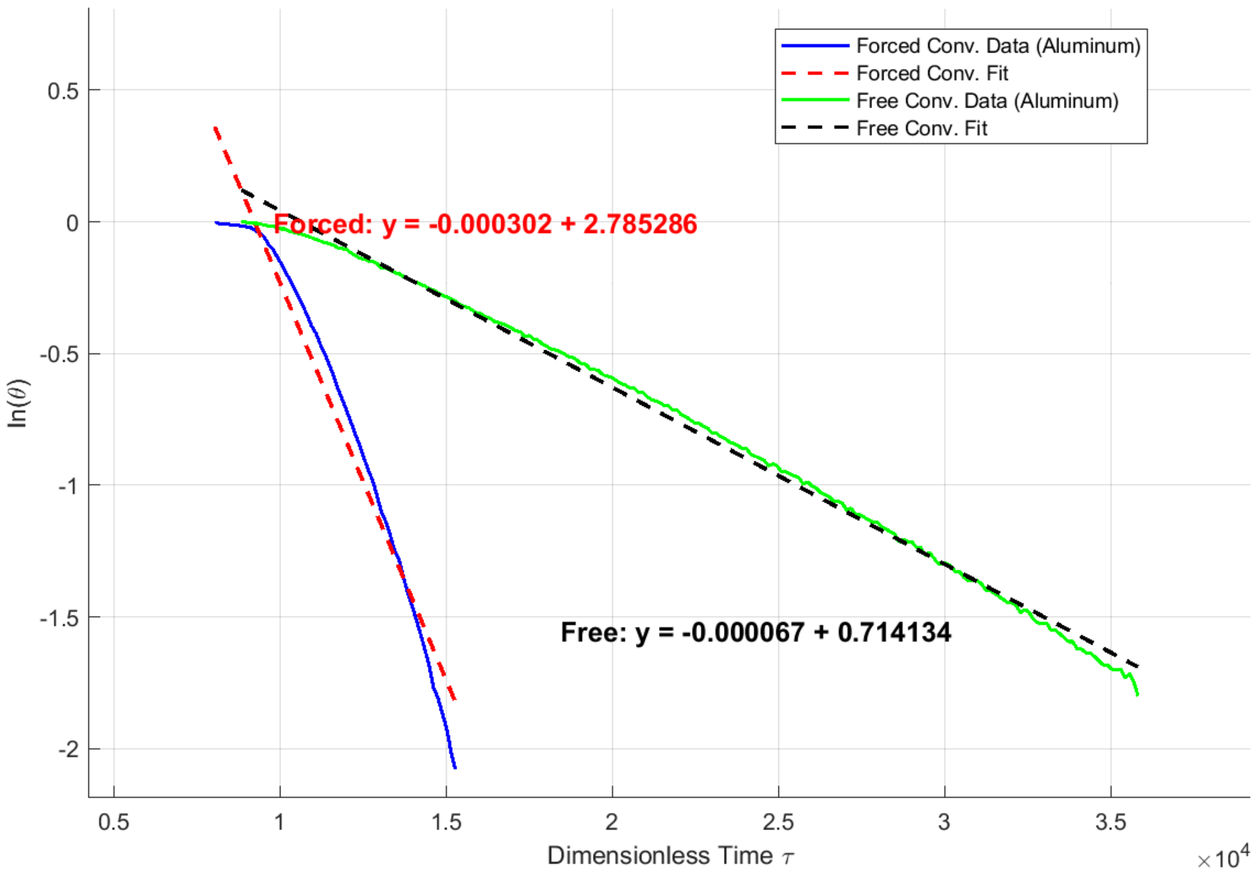

***Figure 2A**. Natural Logarithm of Dimensionless Temperature, ln(θ) plotted against dimensionless time, τ. Least square regression used to extract lines of best fit for both Forced Convection (red dashed line) and Free Convection (black dashed line) for the cooling of an Aluminum square plate.*

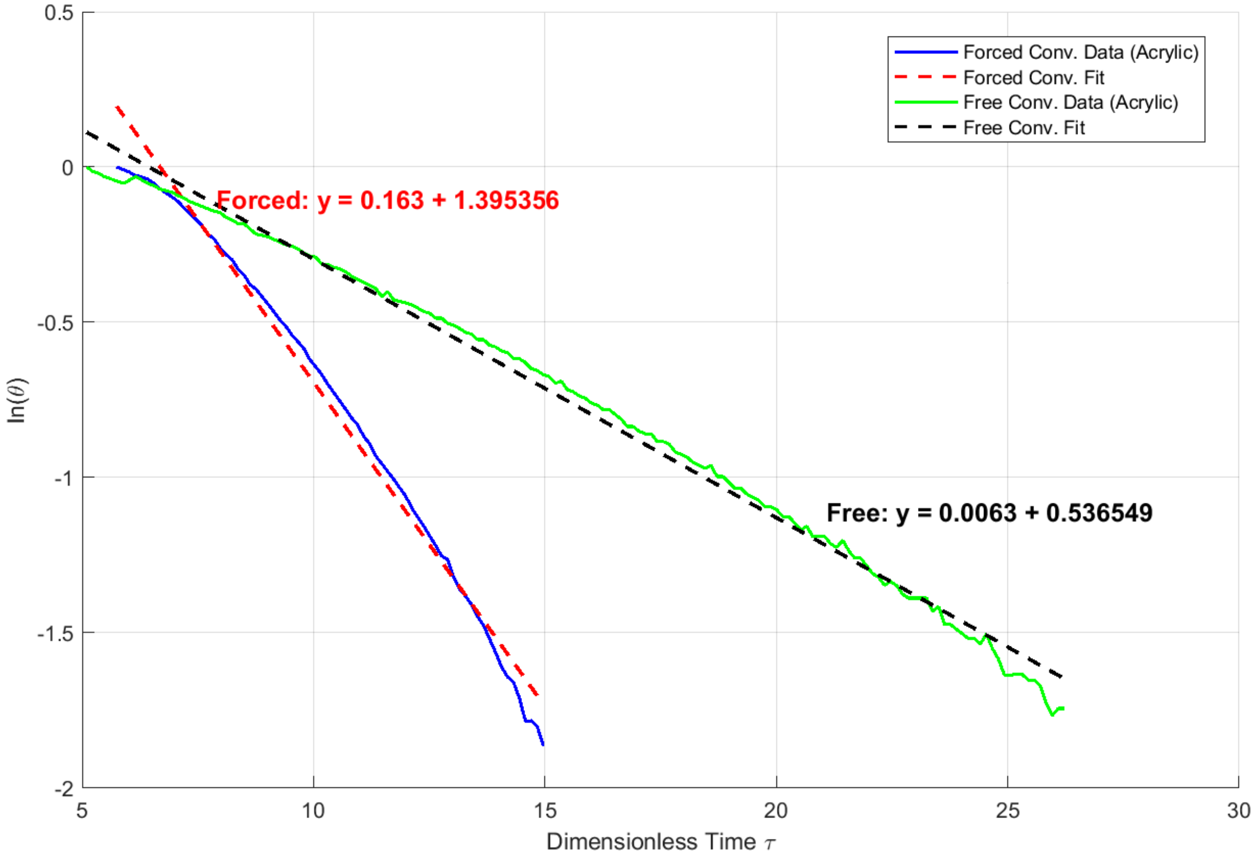

Figure 2B. *Natural Logarithm of Dimensionless Temperature,* $ln(\theta)$ *plotted against dimensionless time,* τ. *Least square regression used to extract lines of best fit for both Forced Convection (red dashed line) and Free Convection (black dashed line) for the cooling of an Acrylic square plate.*

From Figures 2A and 2B, it was evident that the magnitude of the slope for the line of best fit of Acrylic was much larger than that of Aluminum. Additionally, for a given material, the slopes of the forced convection lines were around 3 times greater than the free convection line slopes.

The slopes from the best-fit lines in the plots above were used to experimentally determine the Biot numbers and assess the validity of assumptions made. Comparing Equations 2 and 3, the slope of the curves graphed above is found to be the negative Biot number, as expressed in Equation 4.

$$ln(\theta) = \frac{-t}{\frac{\rho V c}{\bar{h} A}} \quad (3)$$

$$Bi = \frac{\bar{h} L_c}{k} \quad (4)$$

See Appendix derivation section for derivation of Equations (3) and (4).

Material Properties and sample dimensions used to calculate the Biot number, convective heat transfer coefficient, and time constant are located in tables F and G in the Tables section of the Appendix.

Table 1. *Evaluating the experimentally determined Biot Number of Aluminum and Acrylic plates under different convection conditions. Theoretical ranges assumed convective heat transfer coefficient for air is between* 2.5 to $500 W/m^2 K$

Material	Aluminum		Acrylic	
Convection	Free	Forced	Free	Forced
Experimental Biot Number	$0.00007 \pm 9.8 \times 10^{-7}$	$0.00037 \pm 5.18 \times 10^{-6}$	$0.0063 \pm 8.82 \times 10^{-5}$	$0.163 \pm 2.28 \times 10^{-3}$
Theoretical Biot Range	$9.93 \times 10^{-6} - 0.002$		$0.010 - 2.025$	

As shown in the Table above, all but one of the cooling cases result in an experimental Biot number less than 0.1. The Acrylic plate under forced convection was the only outlier, with a Biot number of 0.163.

To experimentally find the convective heat transfer coefficient, h, from the Biot number, equation 4 can be rearranged to form equation 5.

$$\bar{h} = \frac{k Bi}{L_c} \quad (5)$$

Table 2. *Evaluating the experimentally determined convective heat transfer coefficient, h, of Aluminum and Acrylic plates under different convection conditions. Theoretical ranges assumed convective heat transfer coefficient for air is between* 2.5 to 500 W/m^2K

Material	Aluminum		Acrylic	
Convection	Free	Forced	Free	Forced
Experimental convective heat transfer coefficient ($\frac{W}{m^2K}$)	17.63±0.27	92.35±1.41	15.59 ±0.24	40.28±0.62
Convective heat transfer coefficient range ($\frac{W}{m^2K}$)	2.5 - 500		2.5 - 500	

To determine if the experimentally obtained convective heat transfer coefficients were within an expected and acceptable range, a theoretical time constant was found by rearranging equation 3 to find t_c and substituting known values (equation 6). A theoretical range for the time constant was also found by continuing the assumption that the convective heat transfer coefficient of air is between $2.5 - 500 W/m^2K$.

$$\tau = \frac{\bar{h}A}{\rho Vc}t$$

$$\frac{t}{t_c} = \frac{\bar{h}A}{\rho Vc}t$$

$$t_c = \left(\frac{\bar{h}A}{\rho Vc}\right)^{-1} \quad (6)$$

For a square thin plate, it is also known that $\frac{A}{V} = \frac{1}{d}$, where d is the thickness. Therefore, it is apparent that the time constant t_c can be found by equation 6 below. The time constants across the 4 combinations of plate material and convection type are tabulated in Table 3 below.

Table 3. *Evaluating the experimentally determined time constant, t_c, of Aluminum and Acrylic plates under different convection conditions. Theoretical ranges of t_c assuming the convective heat transfer coefficient for air is between* 2.5 to 500 W/m^2K

Material	Aluminum		Acrylic	
Convection	Free	Forced	Free	Forced
Experimental time constant (t_c), (s)	109.64	21.09	122.77	47.52
Theoretical time constant range (t_c), (s)	3.86-770.94		3.82-764.64	

To better understand the range of time constants, Figures 3A and 3B help visualize the temperature decay rates for both materials across forced and free convection.

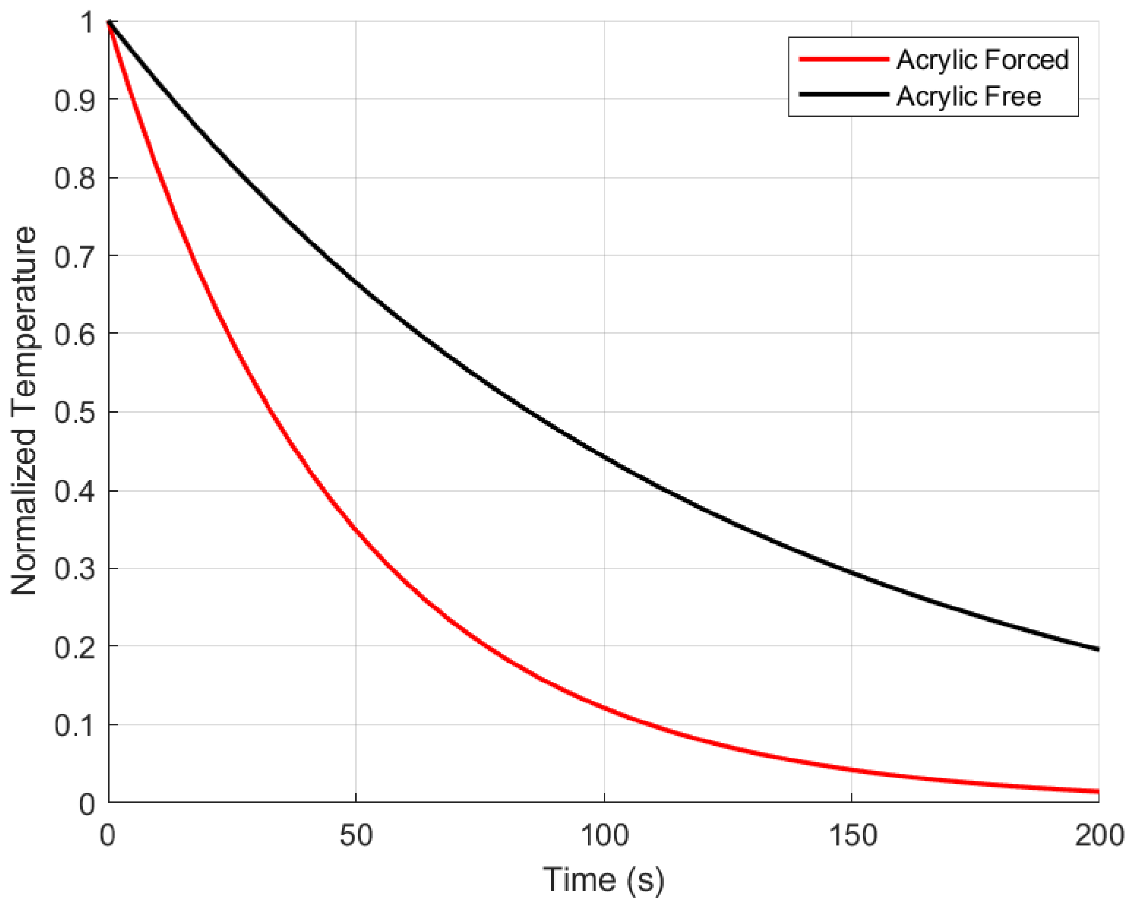

Figure 3A. *Normalized temperature decay across time for Acrylic material under forced (red) and free (black) convection. Curves plotted using experimentally calculated time constants from data for Acrylic experiments and temperature normalized using the highest measured value.*

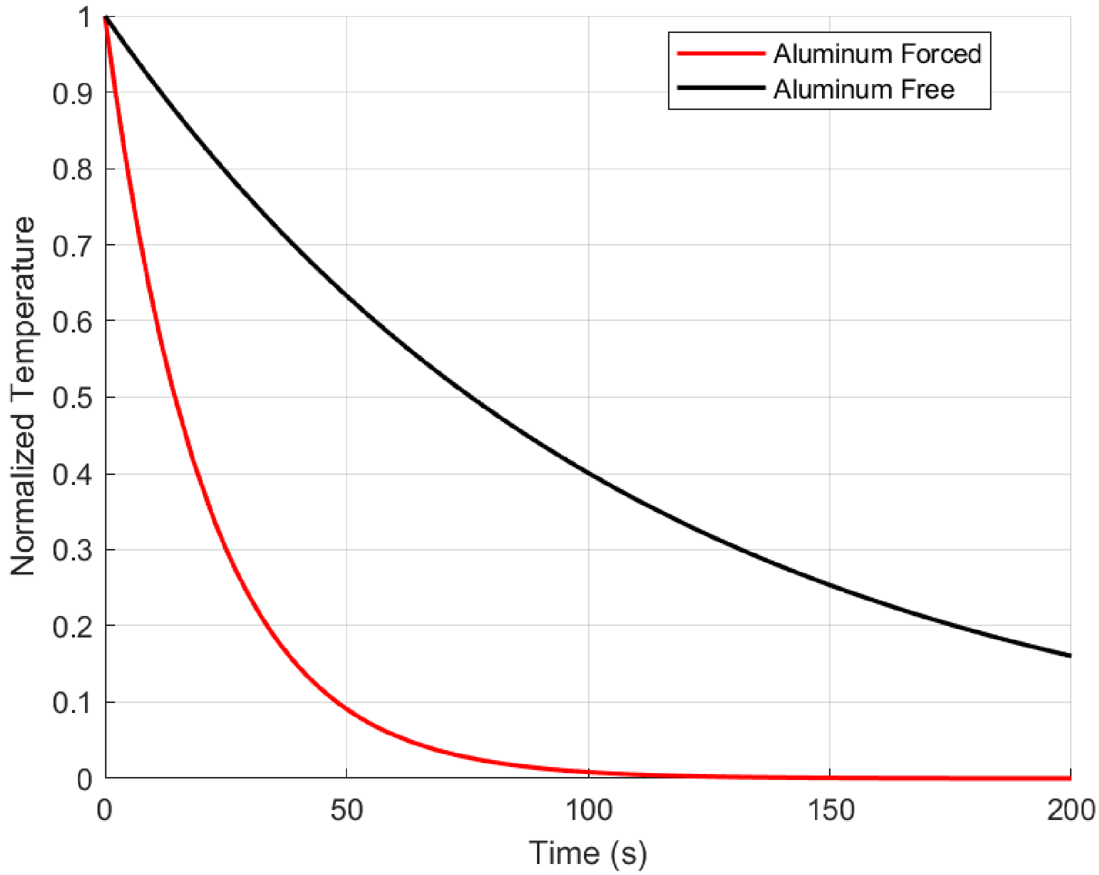

Figure 3B. *Normalized temperature decay across time for Aluminum material under forced (red) and free (black) convection. Curves plotted using experimentally calculated time constants from data for Aluminum experiments and temperature normalized using the highest measured value.*

From the plots, it was observed again that the curves for forced convection had faster decay rates than those for free convection.

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Discussion

The results discussed above were used to analyse trends in the Biot number for different materials and assess the validity of the Lumped Capacitance Model in representing convective heat transfer across a square plate.

Observing higher slopes for forced convection in Figures 2A and 2B, and knowing that they correspond to the Biot number of the material, it can be interpreted that forced convection produced larger Biot numbers. This claim is also evident through the higher convective heat transfer coefficients calculated in Table 2.

Surprisingly, when comparing aluminum and acrylic directly, it was found that the free convective heat transfer coefficient of acrylic (15.59 W/m^2K) was quite close to that of aluminum (17.63 W/m^2K). This was unexpected as the difference in Biot numbers for the two was drastic, and it would be expected that the heat transfer coefficients would follow. On the other hand, the forced convective heat transfer of acrylic (40.28 W/m^2K) was less than half that of aluminum (92.35 W/m^2K), indicating that aluminum had better overall performance when convective heat transfer was induced or forced.

Looking at the Biot number's significance, values below 0.1 show that the internal temperature gradients across the material are negligible, validating the model's assumption of uniform temperature. As indicated by Table 1, most values of the Biot number fell below the 0.1 threshold, indicating that the model was applicable. However, for the case of acrylic under forced convection, its value of 0.163 was well above the 0.1 threshold. This suggests that the lumped capacitance model was not applicable to model the heat transfer for this cooling scenario. To more accurately analyse the heat transfer across the acrylic plate in forced convection, the internal temperature gradients need to be considered, and a more comprehensive model would need to be invoked.

Despite this exception, our experimentally determined Biot numbers, convective heat transfer coefficients, and time constants fell well within the theoretical range of expected values, as indicated by Tables 1, 2, and 3. Table 3 showed that all scenarios experienced had a time constant within our range of expected values.

The calculations in Tables A and B of the appendix, tabulating Biot numbers across all measurements, can be seen to not change much between trials. The consistency of data between trials in both Tables A and B, with generally small standard deviations, suggested that our results were accurate and acceptable for this experiment.

This experiment had many potential sources of error, ranging from thermistor accuracy to response time. These factors can heavily influence temperature readings, especially during shorter cooling periods. Although the Murata thermistors had a tolerance of only $\pm 1\%$, this initially small uncertainty propagated through the numerous calculations, eventually leading to uncertainties of 1.4% for the Biot numbers and 1.53% for the convective heat transfer coefficient. These uncertainties could potentially explain why a Biot number larger than 0.1 was found.

Another major source of potential uncertainty was other modes of heat transfer, such as conduction and radiation. These other heat transfer modes, though ignored by our model, could easily have caused more rapid cooling experienced by the plate and created uncertainty in all of our data and following calculations. Future experiments could mitigate the error caused by these modes of heat transfer by insulating the plate from the ambient environment. This would lower the heat transfer by radiation and conduction and ensure that the primary source of heat transfer would be convection.

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Conclusion

This experiment was able to provide valuable insights into how material properties and convection types affect convective heat transfer rates throughout a material. Our results indicated that, for both materials, forced convection led to higher convective heat transfer coefficients of 92.35 W/m^2K for Aluminum and 40.28 W/m^2K for Acrylic, compared to free convection, which had lower convective heat transfer coefficients of 17.63 W/m^2K for Aluminum and 15.59 W/m^2K for Acrylic. When comparing materials, it was clear that Aluminum exhibited much better performance under forced convection and had only a slight edge compared to Acrylic under free convection.

All the Biot numbers calculated were below 0.1, except for that of the acrylic under forced convection. The Biot numbers being less than 0.1 proved that the Lumped Capacitance Model was an appropriate theory to model the convective heat transfer of the plates. The acrylic under forced convection Biot number being 0.163, larger than 0.1, indicated that the Lumped Capacitance Model was not a suitable model to analyse the convective heat transfer across the plate when experiencing free convection.

For practical uses in science and engineering, Aluminum is a much better choice for applications employing forced convection, such as electrical component cooling and energy generation with coolant flow. An important note was that our experiment focused on the convective heat transfer across a thin square plate, which would mean that other shapes with differing characteristic lengths could easily have different results.

Though in an ideal situation, a metal with extremely high convective heat transfer would improve cooling across most applications, it is important to understand that material properties like density can make systems overweight and impractical. This makes the selection of a material and its properties crucial, as an engineer would aim to maximize cooling efficiency while minimizing weight.

When assessing a square plate under forced conditions, Aluminum easily offers the best convective cooling performance and is the preferred material for all relevant engineering applications. Acrylic can be used as a better alternative for situations where free convection is experienced, as it offers similar performance but comes at a fraction of the cost and is much less dense than Aluminum.

Future studies could build on this experiment by testing more accurate models for heat transfer or by using more realistic testing environments to further studies on material performance and selection.

Appendix

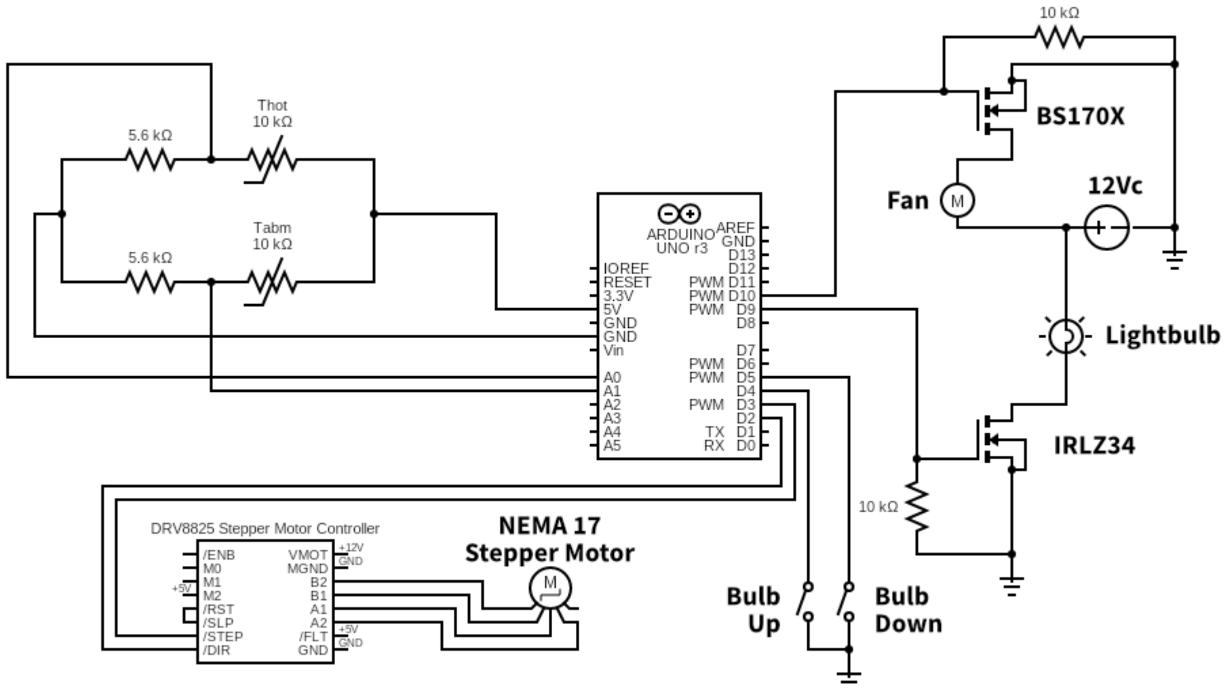

Figure B. *Circuit Diagram used in the experiment*

Equation A:

Standard Deviation

$$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2} \quad (A)$$

σ is standard deviation, N is number of data points, x_i is a single point's value, and $\bar{x}$ is the mean of all data points

Equation B: Error Propagation

$$\text{For } \theta = \frac{X}{Y}\,; \quad \frac{\Delta\theta}{\theta} = \sqrt{\left(\frac{\Delta X}{X}\right)^2 + \left(\frac{\Delta Y}{Y}\right)^2} \text{ (B)}$$

$$\frac{\Delta\theta}{\theta} = \sqrt{\left(\frac{T}{T - T_\infty}\right)^2 + \left(\frac{T_0}{T_0 - T_\infty}\right)^2}\,; \text{ 1\% error in thermistor}$$

$$\frac{\Delta\theta}{\theta} = \sqrt{(0.01)^2 + (0.01)^2}$$

$$\frac{\Delta\theta}{\theta} = 0.014 = 1.4\ \%$$

Assuming this carries through plotting and slope derivation, results in 1.4% uncertainty in the Biot number

$$\frac{\Delta B}{B} = 1.4\ \%$$

Applying Equation B to Equation 3 derives the error propagation formula for the convective heat transfer coefficient below, treating the thermal conductivity, k, as a constant

$$\frac{\Delta h}{h} = \sqrt{\left(\frac{\Delta B}{B}\right)^2 + \left(\frac{\Delta L}{L}\right)^2}$$

Measurements of thickness and width were made with a precision of 0.01 mm, so the uncertainty follows as half of the smallest unit. The characteristic length is drawn from the thickness, d, so it can be used as an estimate.

$$\Delta L = \frac{0.01}{2} = 0.005 \text{ mm}$$

$$\frac{\Delta L}{L} = \frac{\Delta d}{d} = \frac{0.005}{0.81} = 0.0062 = 0.62\ \%$$

Combining these values, the uncertainty for h is found below:

$$\frac{\Delta h}{h} = \sqrt{(0.014)^2 + (0.0062)^2}$$

$$\frac{\Delta h}{h} = \sqrt{(0.014)^2 + (0.0062)^2}$$

$$\frac{\Delta h}{h} = 0.0153 = 1.53\%$$

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Lumped Capacitance Model Equation Derivations

To derive equations (3) and (4), the Lumped Capacitance Model is used with an energy balance:

$$\rho V c \frac{dT}{dt} = -\ hA(T\ - T_{\infty})$$

First, the variables are separated:

$$\frac{1}{T - T_{\infty}}\ dT\ = \frac{-hA}{\rho V c} dt$$

By then integrating from $t\ =\ t_0$ ($T\ =\ T_{\infty}$) to time t, the equation becomes:

$$ln(\frac{T - T_{\infty}}{T_0 - T_{\infty}}) = -\ \frac{hA}{\rho V c} t\ (a)$$

And by defining the dimensionless temperature parameter $\theta(t)$:

$$\theta(t)\ = \frac{T - T_{\infty}}{T_0 - T_{\infty}}$$

Then equation (a) becomes equation (3):

$$ln(\theta)\ = -\ \frac{hA}{\rho V c} t\ \ (3)$$

A time constant is then defined τ_c:

$$\tau_c\ = \frac{\rho V c}{hA}$$

Such that (b) becomes:

$$ln(\theta)\ = -\ \frac{t}{\tau_c}$$

To derive the Biot number Equation (4), the conduction and convection resistance equations are defined as:

$$R_{cond}\ = \frac{L_c}{kA},\ R_{conv}\ = \frac{1}{hA}$$

By then taking the ratio of R_{cond} to R_{conv}, the Biot number and equation 3 can be derived:

$$Bi\ = \frac{R_{cond}}{R_{conv}} = \frac{\frac{L_c}{kA}}{\frac{1}{hA}}$$

$$Bi\ \ = \frac{hL_c}{k}\ (4)$$

Appendix Tables

Table A. *Calculating Biot numbers for Acrylic and Aluminum plates under forced and free convection types over 3 trials, and finding the standard deviation*

Material	Aluminum Biot		Acrylic Biot	
Convection Type	Free	Forced	Free	Forced
Trial 1	0.000065	0.00032	0.00819	0.0212
Trial 2	0.000071	0.00037	0.0890	0.221
Trial 3	0.000074	0.00041	0.0869	0.232
Standard Deviation	0.0000043	0.000042	0.0027	0.0076
Mean	0.00007	0.00037	0.0063	0.163

Table B. *Calculating convective heat transfer coefficients for Acrylic and Aluminum plates under forced and free convection types over 3 trials, and finding the standard deviation*

Material	Aluminum $\overline{h}$ (W/m^2K)		Acrylic $\overline{h}$ (W/m^2K)	
Convection Type	Free	Forced	Free	Forced
Trial 1	16.37	80.59	14.85	38.51
Trial 2	17.88	93.19	16.16	40.10
Trial 3	18.64	103.26	15.76	42.25
Standard Deviation	1.15	11.36	0.668	1.88
Mean	17.63	92.35	15.59	40.28

Table C. *Calculating the time constants for Acrylic and Aluminum plates under forced and free convection types over 3 trials and finding the standard deviation*

Material	Aluminum $\overline{h}$ (W/m^2K)		Acrylic $\overline{h}$ (W/m^2K)	
Convection Type	Free	Forced	Free	Forced
Trial 1	117.74	23.92	128.73	49.64
Trial 2	107.79	20.68	118.29	47.67
Trial 3	103.40	18.67	121.29	45.24
Standard Deviation	6.00	2.16	4.39	1.80
Mean	109.64	21.09	122.77	47.52

Table D. Equation 1 Variable Definitions

T = current temperature
T_0 = initial temperature
T_∞ = the ambient temperature.

Table E. Equation 2 Variable Definitions

k = thermal conductivity	A = plate area
t = time	ρ = plate density
c = specific heat of the material	d = thickness
V = volume	

Table F. Material Properties

	ρ (kg/m^3)	$c\ (J/Kg\ K)$	$k\ (W/mK)$
Acrylic	2707	879	204
Aluminum	1180	2000	0.2

Table G. Sample Dimensions

$d = 0.081\times10^{-3}$ m
$A = 1.45161\times10^{-3}\ m^2$

Bibliography

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(4) *"Heat Transfer Fundamentals: Conduction, Convection, Radiation." (Mechanical Engineers' Handbook, Energy & Power, Vol. 4, 3rd ed.) by G. P. Peterson. — Free PDF page for relevant chapter: https://studylib.net/doc/18842081/chapter-5-heat-transfer-fundamentals*

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Critiques

Sainirnay's Critiques

Andrew Cagney's Conclusion

The conclusion does a good job starting by summarizing the key motivations behind this experiment. Key findings of the experiment are stated, but quantitative data and numbers found in the experiment are not mentioned. Since it is the conclusion, a conclusive summary of what was found in the experiment would serve well in this section. Important takeaways are well described, though, and it is clear to see why this experiment was useful. Real-world implications are also well discussed, tying back to what was mentioned in the introduction. The ending could be improved by suggesting potential improvements on the experiment itself or any additional tests that could be performed following this data. For instance, a follow-up experiment could be suggested that uses these materials as heat sinks in actual electronic systems to weigh their performance. Overall, this conclusion does an excellent job of summarizing key findings of the lab, but could be improved by expanding on how this lab can be built on.

Leila Juan's Introduction

This introduction is very well written and does an excellent job of following the rubric. The introduction to the topic of the experiment has the appropriately sized scope and is concise enough to understand what is set up in the rest of the experiment. The claims made are also backed by a reference that the author cites. One improvement might be to include one or two more references for showcasing why this topic of study is important or what other studies are being carried out in this field. After setting up this background, the knowledge gap is set up well to naturally follow from the previous conversation. This is done in a single sentence, but it is quite long and reads more like a run-on sentence. Another improvement would be to either make the statement more concise or split it up into a couple of sentences with the intended detail. The knowledge gap is followed by a description of the experiment in a way that feels responsive, which again follows the rubric. The summary of what was done in the experiment is well written, being detailed enough to understand the context without being overly descriptive. However, some terms like the h value and Biot number are not exactly defined, which may be hard for readers to follow. If they could be described similarly to how "(t)" is defined, without going into too much theory, the experiment may be easier to follow. Overall, this introduction meets the requirements of the rubric and could be improved with a couple of improvements on structure and context.

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Owen Hanenian's Abstract

This abstract does a good job of following the main parts of the rubric, but could use some more elaboration on some points. It starts with a brief introduction to the topic, but it feels somewhat vague, mentioning that heat transfer is important in engineering. Tailoring this to a more specific engineering application in the first sentence could better set up the historical background. The second sentence concisely states what the work is about and follows the rubric by not going into too much detail. The approaches and methods used are also stated in a way that is easy to follow and brief, which is well fit for an abstract. One improvement might be to explain a little bit about what forced and free convection are, so people outside the discipline can follow along as well. The abstract concludes by summarizing the most important results and their implications, but the ending sentence is a bit confusing. The tie to Artificial Intelligence does not flow with what is being set up about heat transfer. A more direct application of what these results can be used to achieve would strengthen the conclusion. Overall, the abstract follows the rubric very well, but could use more focus in the opening and concluding sentences.

Gregorio's Critiques

Ting Wang's abstract

The abstract does a good job of providing a brief introduction of the experiment with a concise statement of the phenomenon studied and a main summary of the most important results was given. The content of the experiment is summarized well enough such that it can be understood without reading the main content. This abstract does a great job of discussing the main methodology and explains the important results, however it would be beneficial to adapt the abstract such that a scientists/engineer from another field could understand the experiment, possibly by providing brief explanations on key terms like: lumped capacitance model, Biot number, convective heat transfer coefficient. This abstract could also benefit from tying this experiment to broader scientific applications with specific examples as to how the experiment's findings could impact a variety of fields such as electronic component cooling, energy production and so on. Additionally, some sentences have several grammatical errors that hinder the fluidity and structure of the abstract, though a minor issue, the abstract would greatly benefit from having these corrected to improve the fluidity of the reader's reading experience.

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MAE 170 Lab 7

Allen Vadukoot's Introduction

This introduction has a great opening sentence that directly ties into the experiment, the importance of the experiment and why this phenomenon is being studied. The introduction uses a broad scope of examples, but not too broad, to explain the importance of the work such that any reader, even a scientist/engineer from another field would have an easy time understanding. The problem statement is clear and directly to the point, allowing the reader to understand the aim of studying this phenomenon. This is followed by a comprehensive statement describing what work was done, and responding to the problem statement. The main downfall of the introduction is that it lacks in text references to support the various claims made, though not a major issue, this would allow for a better reading experience as an interested reader, from any field, could be allowed to read more into the phenomenon of this experiment by linking references to the content being discussed.

Anvita Nandyala's Results & Discussion

Anvita's Results section employs many figures and tables, all of which are well labeled and easy to understand such that conclusions can easily be drawn by a prospective reader. The results section makes good use of structure and section labelling to guide the reader as to what part of the experiment is being discussed and analysed. Each section makes use of figures and tables, while following this with an appropriate discussion of the key results, trends and observations made in each section of the experiment. The calculations of key results are well presented and easy-to-follow, however, it is not clear where the uncertainty values for each calculation come from, thus this report would greatly benefit from some hindsight as to how the uncertainties were calculated. The discussion discusses key results in context with the phenomenon being investigated and how uncertainties can play a role in skewing data very effectively. Future improvements on this experiment are well explained and brought into the discussion clearly. Though trends and findings are clearly explained to the reader, this section would benefit from using interpretive language rather than direct language, for instance using sentences like "this suggests," or "one possible explanation could be," as this would emphasise to the reader, that there is no one comprehensive correct answer, but rather there can always be many possible reasons to explain the observed behavior.